RC CAR

PROJECT NAME:	Design And Development Of RC Car
PROJECT GROUP:	Deep Goje, Yuraj Harne, Tejas Kale

WHY THIS PROJECT ?

- To learn and experiment with designing a custom RC chassis using aluminium pipes.
- To gain hands-on experience in mechanical assembly and material handling.
- To prepare for RC competitions by testing stability, performance, and durability.
- To encourage creativity and innovation by building something ourselves instead of using pre-made parts.

INTRODUCTION

- Designed and built a custom RC chassis using aluminium pipes.
- Explored a new type of chassis made entirely by ourselves.
- Learned about mechanical design, material selection, and structural stability.
- Prepared for RC competitions by testing performance and durability.

KEY FEATURES

- Custom Aluminium Chassis: Built from scratch for strength and lightweight design.
- Durable Structure: Aluminium pipes provide stability during RC operation.
- Modular Design: Easy to assemble, modify, or repair for experiments or competitions.
- Competition-Ready: Designed to perform in RC racing or obstacle challenges.
- Hands-On Learning: Focused on practical mechanical skills and material handling.

ARCHITECTURE

- Metallic Frame: Custom aluminium chassis forming the main structure.
- BLDC Motors: Provide efficient and high-speed propulsion for the RC vehicle.
- Transmitter & Receiver: Enable remote control communication for movement and navigation.
- Power Distribution Board: Distributes electrical power efficiently to all components.
- ESC & LiPo Battery: Electronic Speed Controllers regulate motor speed; LiPo battery supplies high-energy power.

RESULTS

- Successfully designed and built a custom RC chassis using aluminium pipes.
- Vehicle was stable, durable, and functional for RC operation.
- Gained hands-on experience in mechanical design, assembly, and electronics integration.
- Tested in RC competition scenarios, validating performance and modular design.

FUTURE SCOPE

1. ☐ **Explore lightweight composites or alloys for better performance.**
2. ☐ **Add sensors for autonomous navigation or obstacle detection.**
3. ☐ **Upgrade motors, ESC, or battery for higher speed and efficiency.**